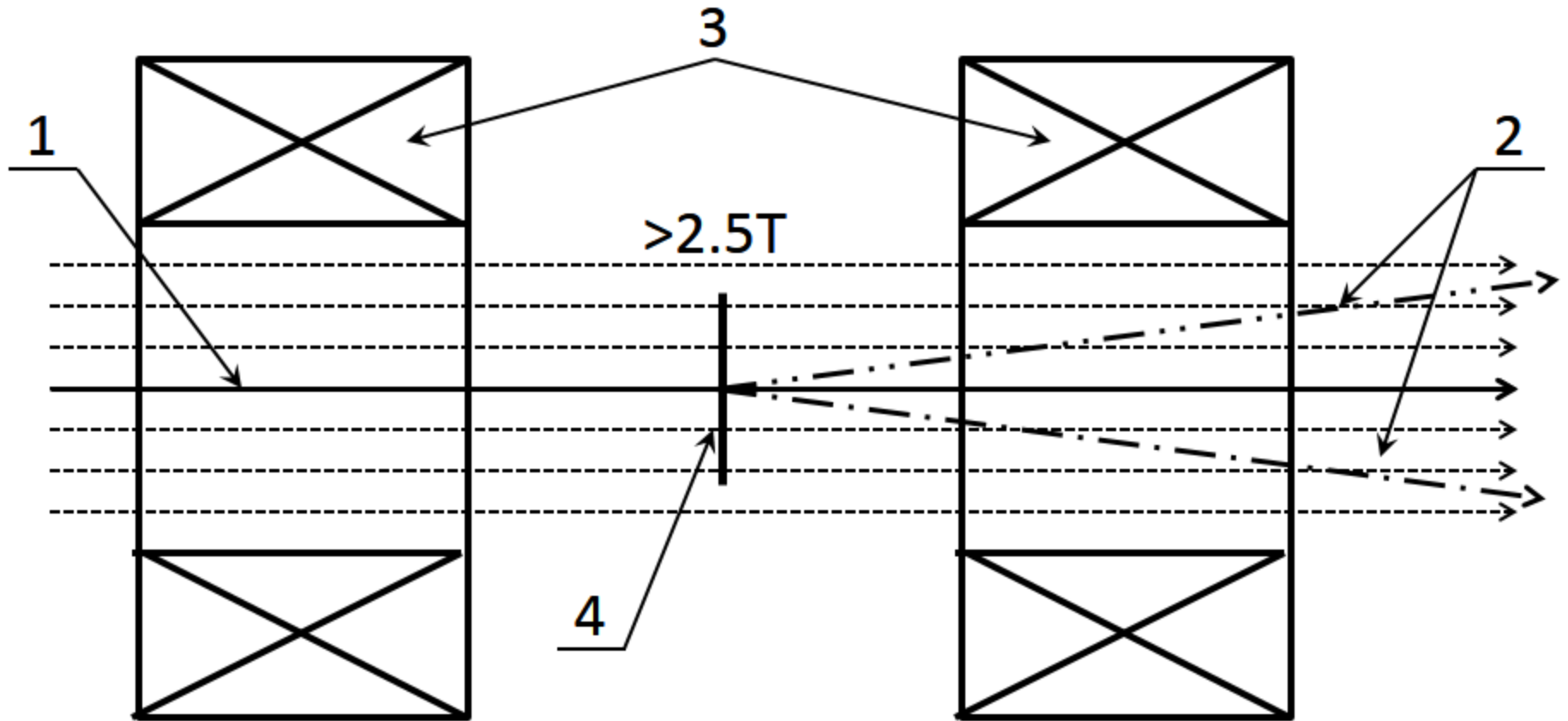




Rensselaer

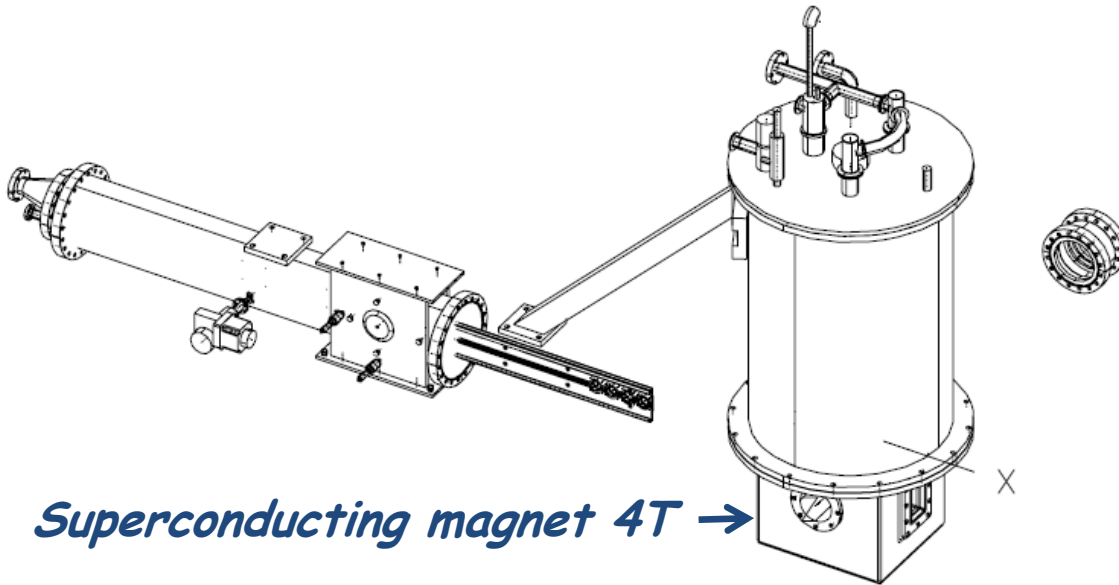
Welcome!

Brute Force

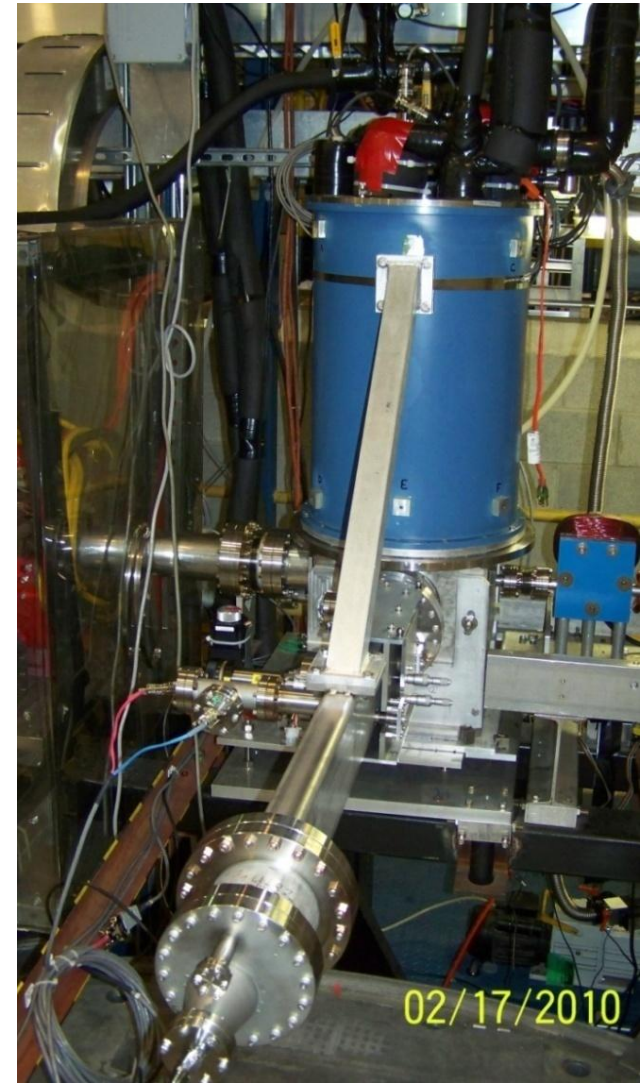
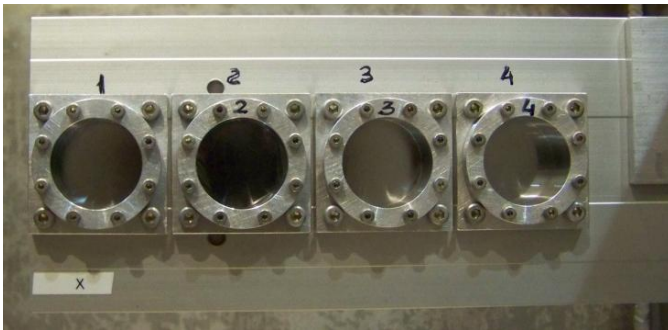
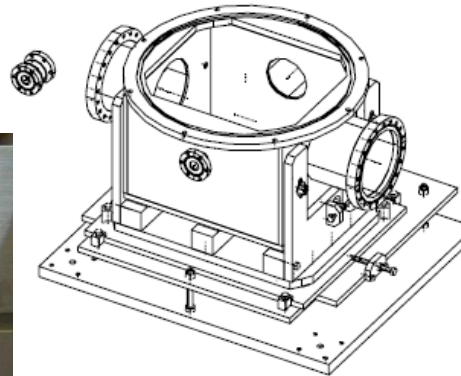


1 – electron beam, 2 – Møller electrons, 3 – magnetized coils,
4 - target

“High Field” Target



Superconducting magnet 4T →

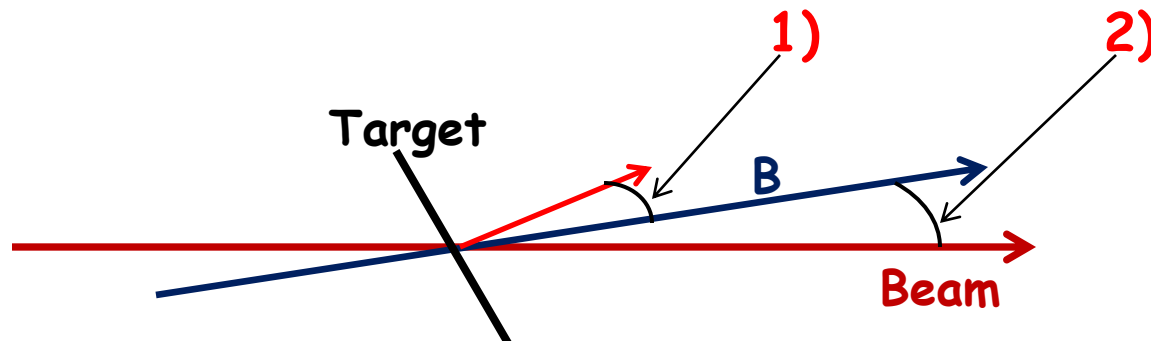


Targets: pure Fe foils 1-1 μ m, 2-4 μ m, 3-4 μ m, 4-10 μ m

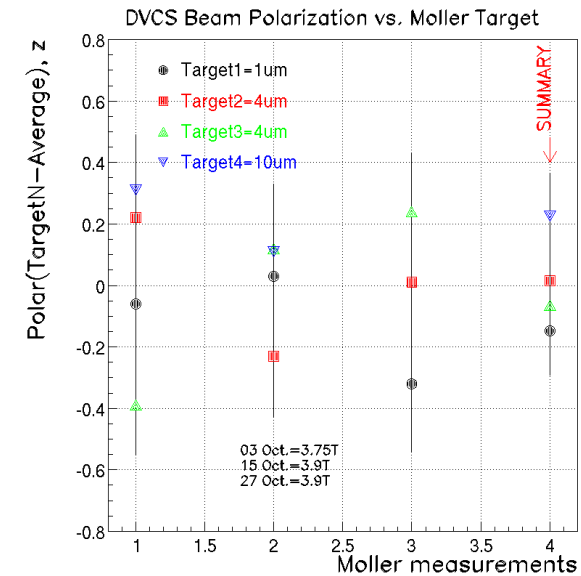
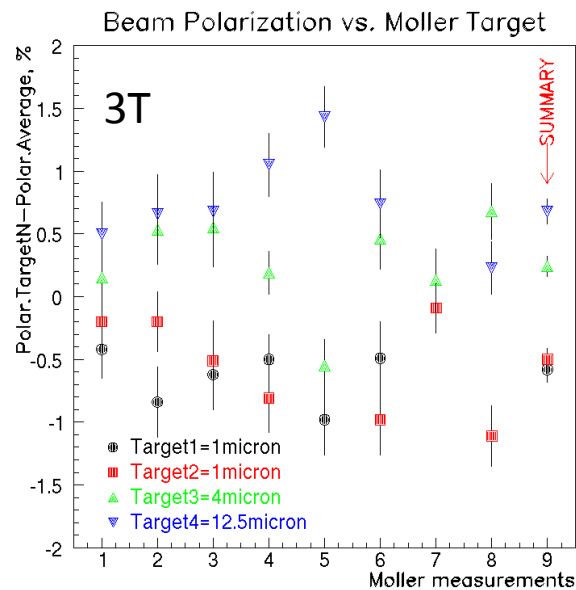
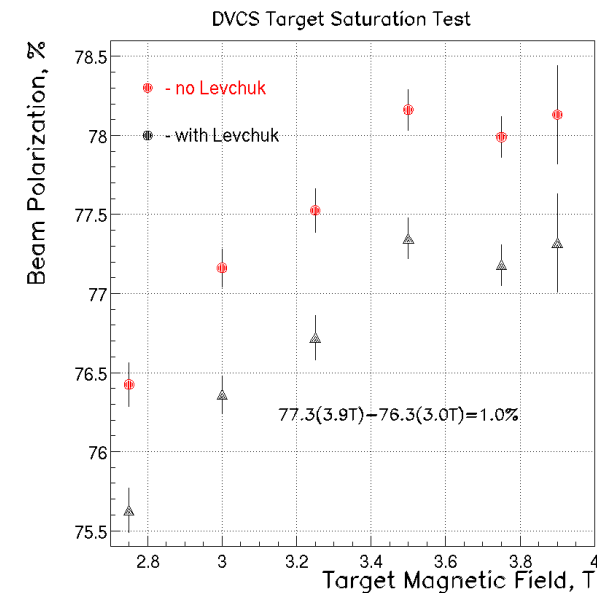
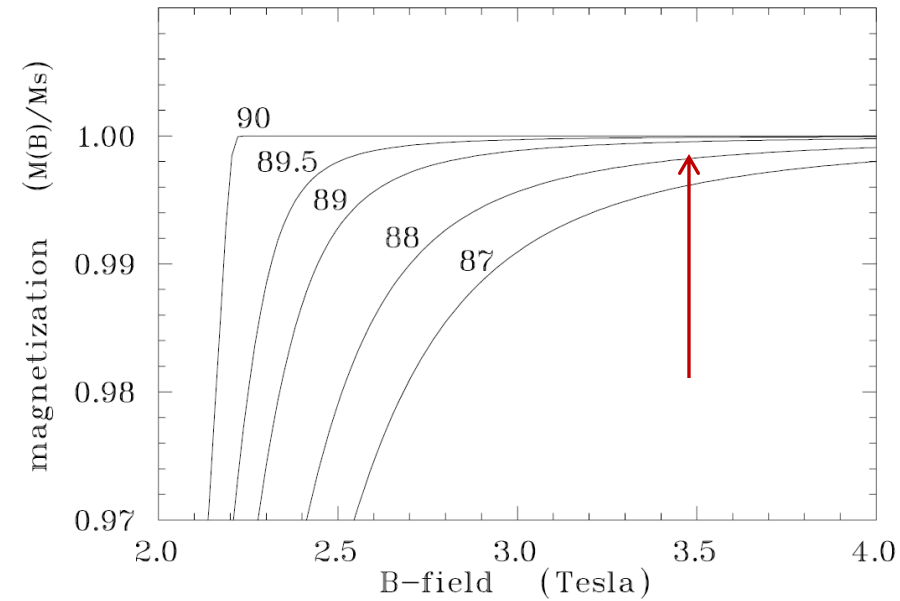
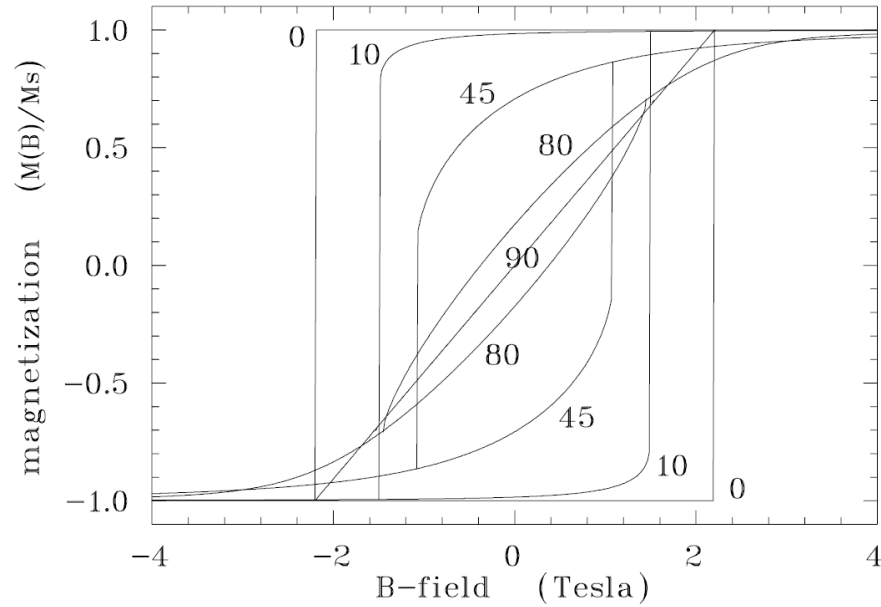
Alignments

Two alignments is a key issue for the “strong field” target:

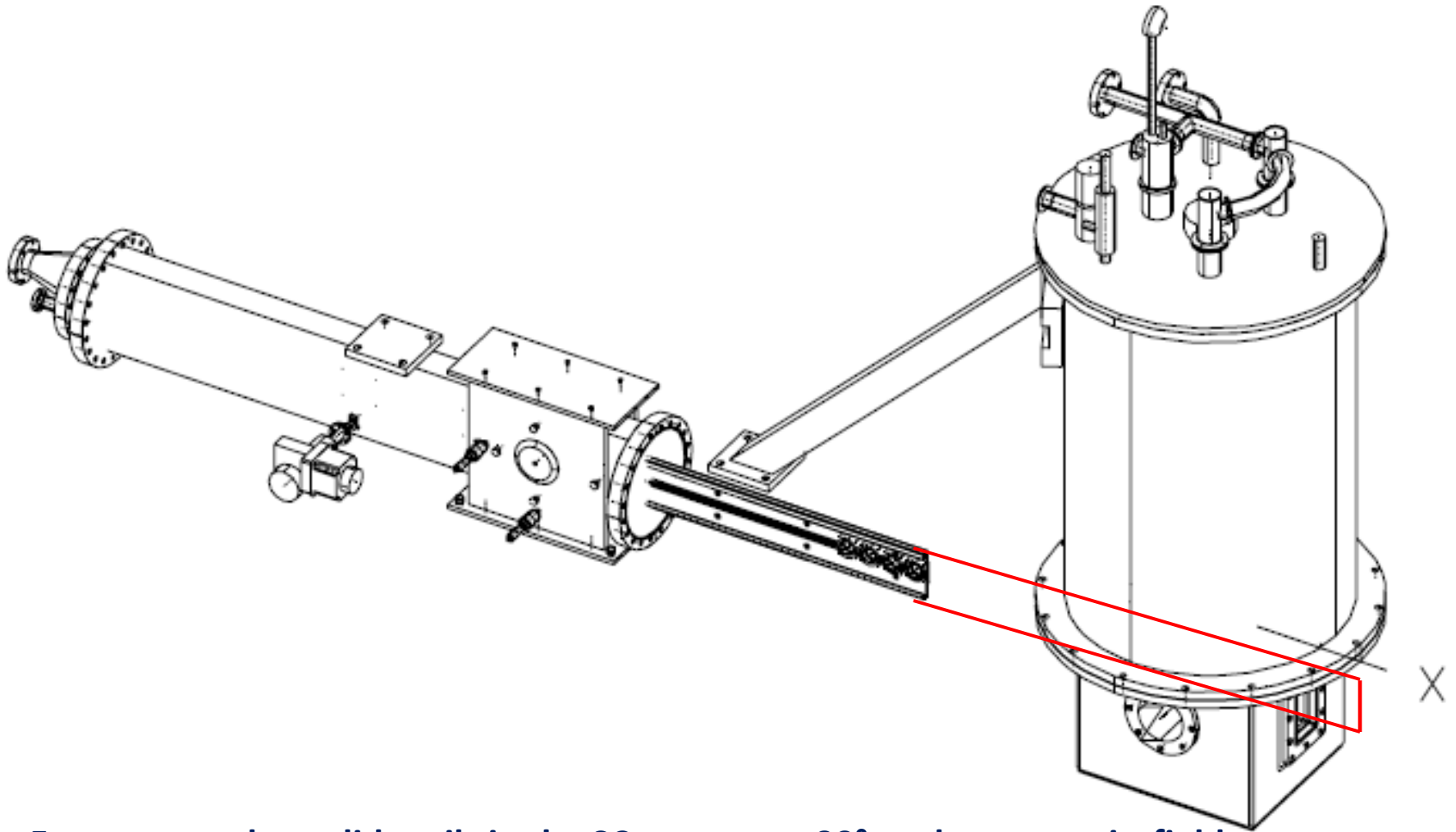
- 1) **Target vs. magnetic field** (saturation, discrepancy between targets)
better alignment or/and stronger magnetic field
Test: foils saturation, measured polarization vs foil #
- 2) **Magnetic field vs. beam direction** (analyzing power, Levchuk-effect, wrong polarization)
better alignment
Test: beam deflection (BPMs downstream of the polarimeter)



Target vs Magnetic field



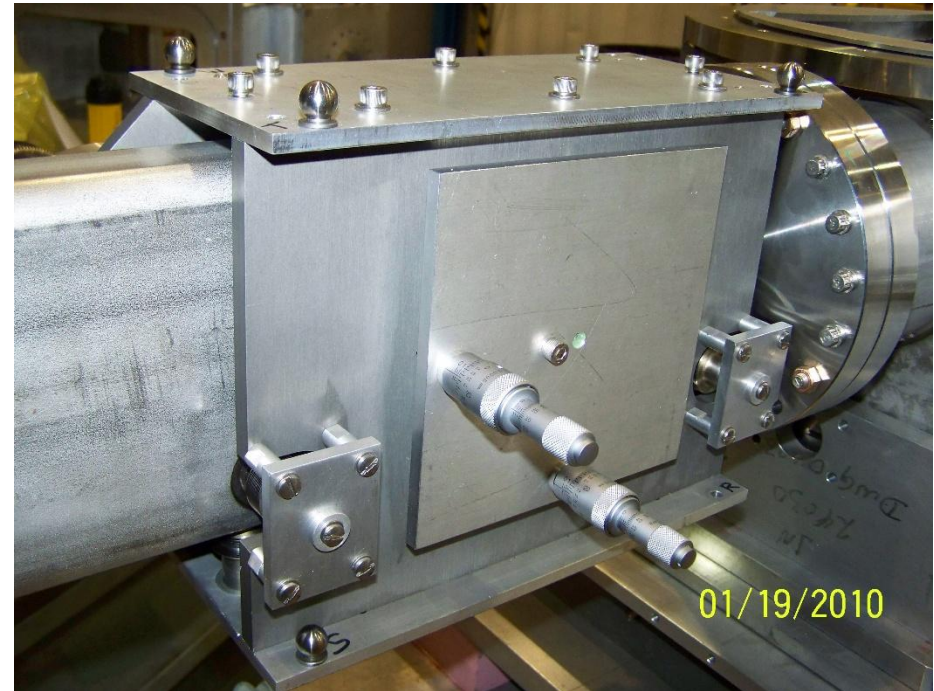
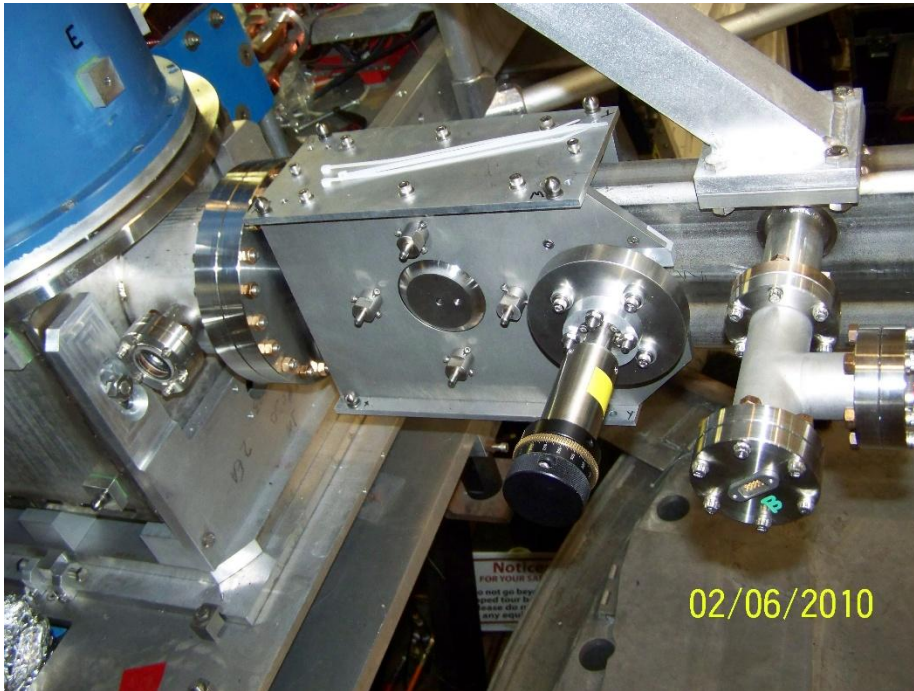
Target vs Magnetic field



- 1) Frame-extender + slide rails in the SC magnet at 90° to the magnetic field
- 2) Power supply test (after repair)
- 3) New magnet field mapping (field mapper restored/new?)

Target vs Magnetic field

Ed Folts



Restoration of adjustment mechanism

- 1) Bellows clean up and welding to fix vacuum leaks**
- 2) If it does not work:**
 - cut bellows and weld another bellows**

**Adjustment + alignment with no magnetic field
to prove mechanism and target loader conditions**

Magnetic field vs. Beam

1) Alignments results

March 30, 2010: $dX=-0.3\text{mm}$ $dY=-1.1\text{mm}$ $d\text{Horiz}=-1.8\text{mrad}$ $d\text{Vert}=+4.4\text{mrad}$

April 16 (Surv.): $dX=-0.4\text{mm}$ $dY=+0.2\text{mm}$ $d\text{Horiz}=+0.7\text{mrad}$ $d\text{Vert}=-1.9\text{mrad}$

April 17, 2010: $dX=-0.2\text{mm}$ $dY=-1.7\text{mm}$ $d\text{Horiz}=-1.6\text{mrad}$ $d\text{Vert}=+6.1\text{mrad}$

July 15 (Surv.) $dX=-0.4\text{mm}$ $dY=+0.1\text{mm}$ $d\text{Horis}=+0.7\text{mrad}$ $d\text{Vert}=+1.9\text{mrad}$

Sept. 30, 2010: $dX=-0.2\text{mm}$ $dY=-1.0\text{mm}$ $d\text{Horiz}=-0.2\text{mrad}$ $d\text{Vert}=+2.9\text{mrad}$

We don't know magnetic field axe position and behavior!

Low field target is seating 0.3 mm upper of the beam center

Cartridges between beam line frame and the target bench are too long

Cut/order new cartridges?

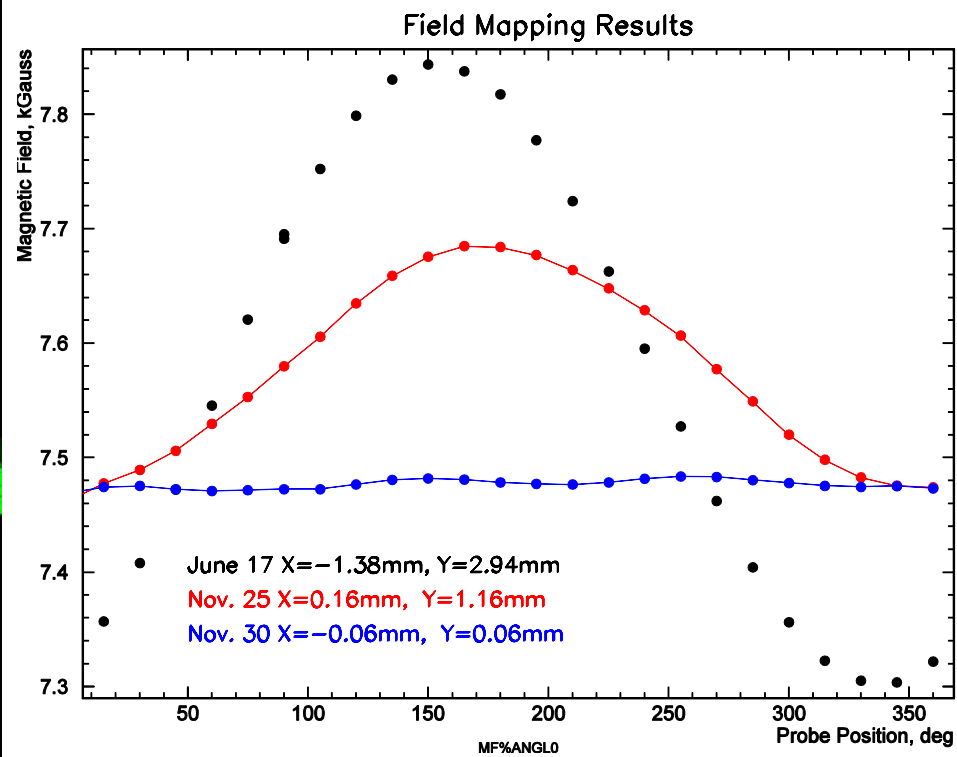
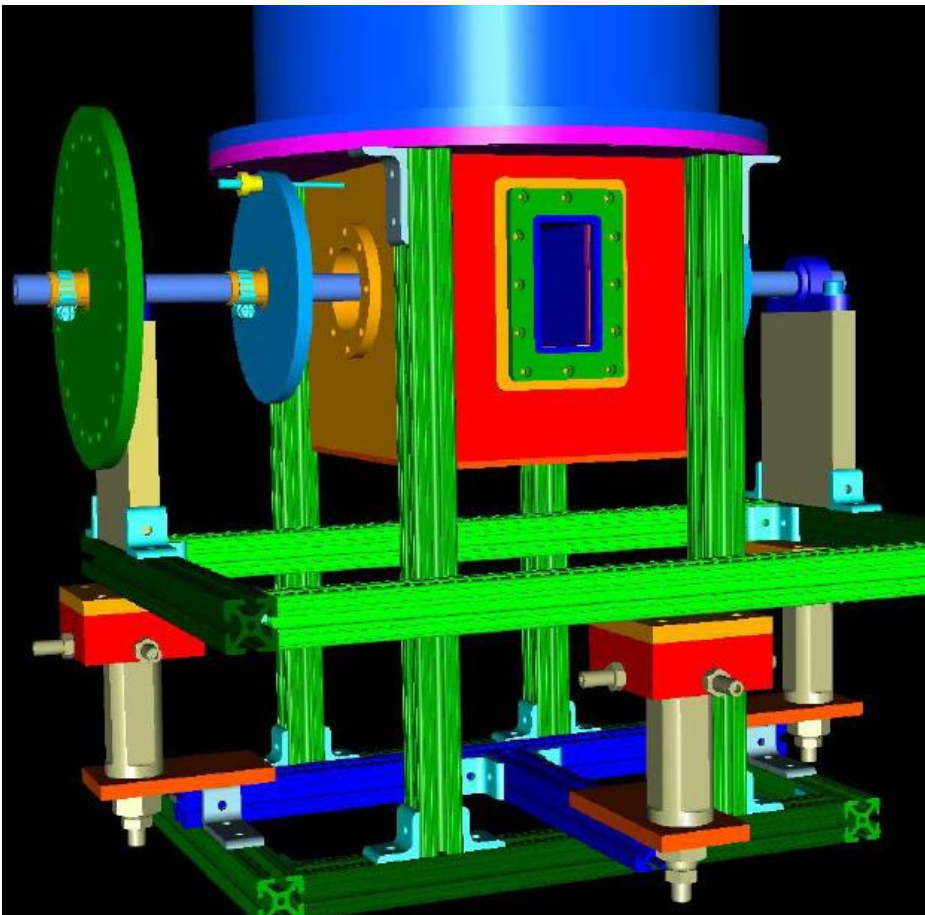
We can run low field target as is.

Fix the cartridges before the high field target installation.

2) New SC magnet alignment (with new field mapping results)

3) Alignment test on the beam (beam deflection on BCMs)

Old field mapper



New field mapping

Field map measurement on the beam line:

- includes iron scattered fields and fringe fields
- new field mapper is needed

Field mapper - alternative

Sasha,

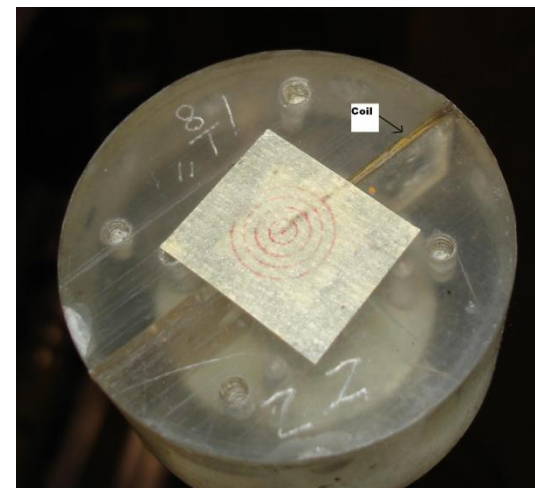
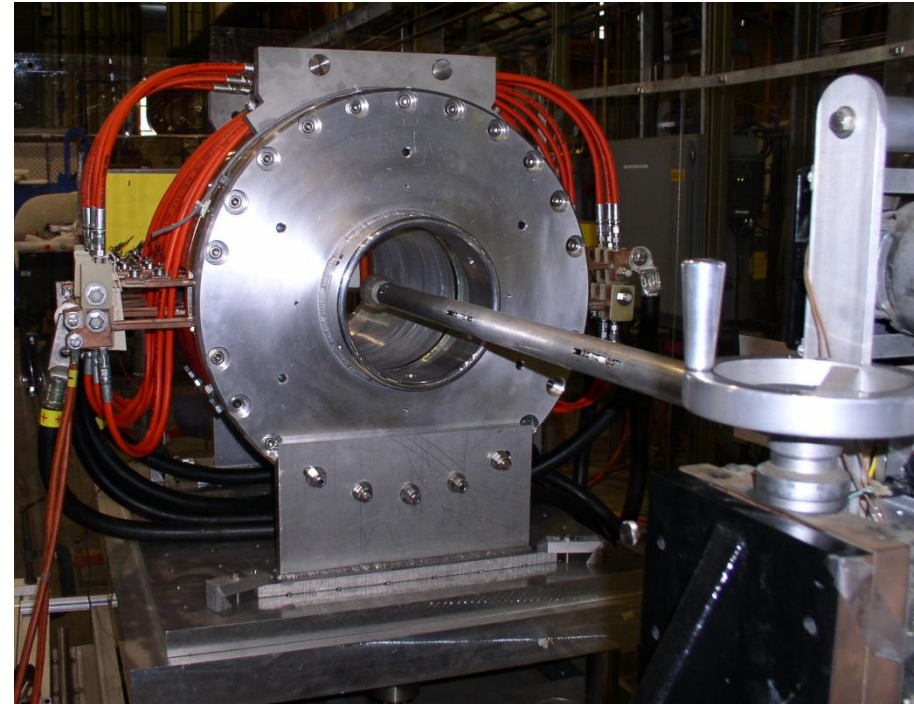
I received this information from my colleague at SLAC concerning solenoid alignment. It seems like a fairly straightforward method although we would have to build something to do this here. It would probably be possible to "wind" the coil using a circuit board instead of wire.

cheers,

Ken

For solenoids, we use a short rotating coil on a long rotating shaft (see pict) to do the alignment and measure the dipole and first few harmonics. Our coils rotate at between 10-25 Hz and their signals are readout with a lock-in amplifier (Stanford Research SR850) that is triggered by a pulse coming from the rotation setup (an encoder). The lock-in reads the voltage from the coil for each harmonic and the B field is related to the voltage by $B = V/(r * w * L * N)$, where r is the radius of the coil, w is the rotational frequency ($2 * \pi * \text{speed in Hz}$), L is the length of the coil, N is the number of turns in the coil. Since r and L are not always known for a hand wound coil, we calibrate the coil constant K ($K = 1/(r * w * L * N)$) by putting it in a dipole (uniform) field, B , which we measure using a hall probe and then solving for K using the voltage from the lock in.

To align the solenoid in X and Y, one puts the coil at one end of the solenoid where the field is bending, at the pole piece usually, and centers the coil until the signal on the lock-in for the 1st harmonic (dipole) is zero. This is the center for that end. Do the same thing on the other end and find the other center. The center of the rotational axis of the coil is where the center is located. Our coils have targets on the front face that our alignment people can read. (see picture Short Coil with text)



Cryogenics line

Final design

Milestone

Money

To do list

Meetings with:

Hall A (Javier)

Kharkov (Sasha)

RPI (Jim)

Hall A techs (Ed)

Hall A engineers (Al)

Jlab alignment group (Chris)

To find final decision with:

the target

field mapper

milestone